

ES -116

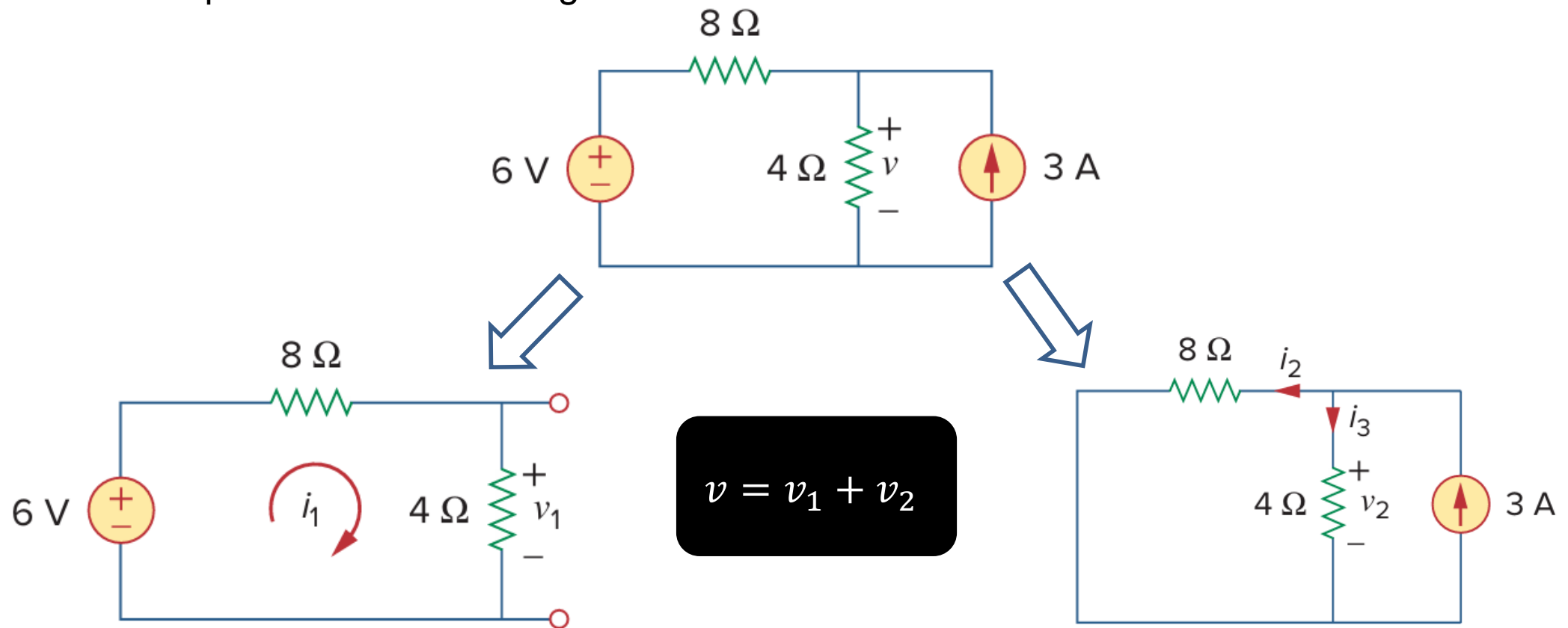
Principles and Applications of Electrical Engineering

Semester II, 2025-26

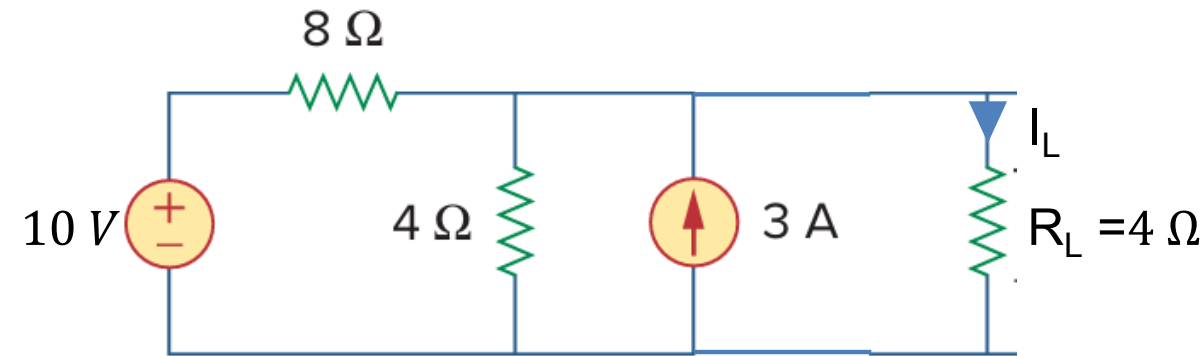
Additional Analysis Techniques -II

Recap: Superposition Method

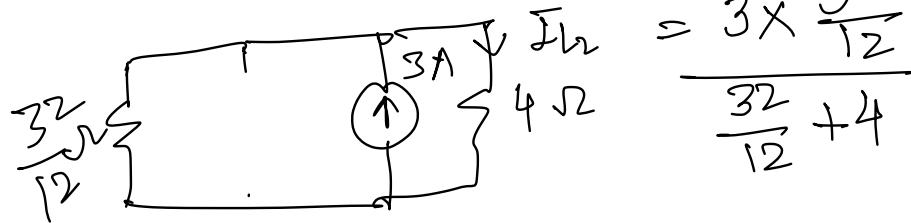
→ The superposition principle states that the voltage across (or current through) an element in a **linear circuit** is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.



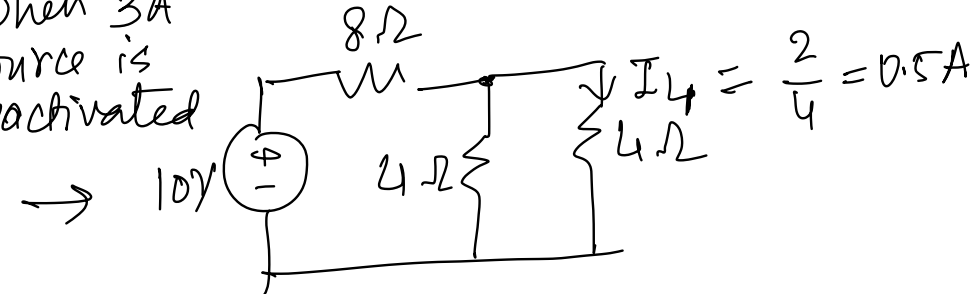
Question (Find I_L)



When 10V source is deactivated :-



When 3A source is deactivated

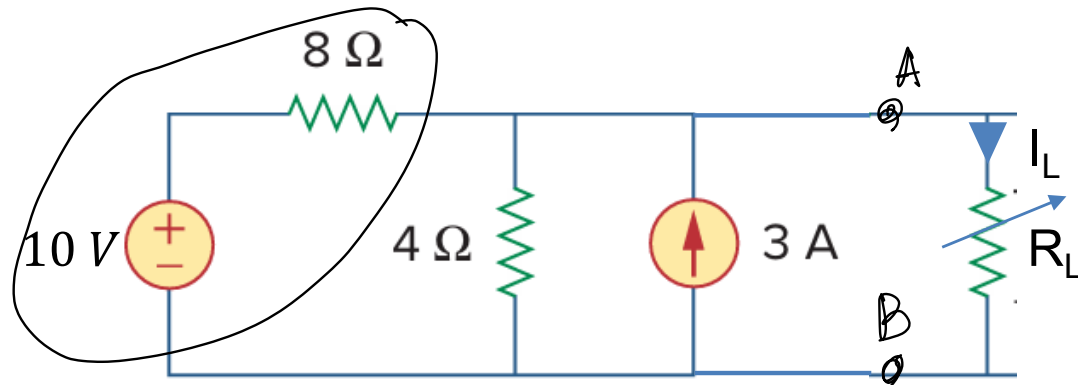


Using superposition:-

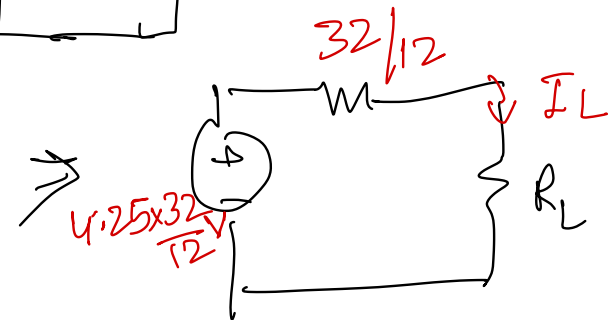
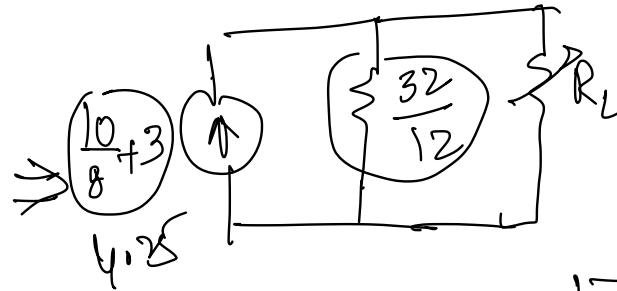
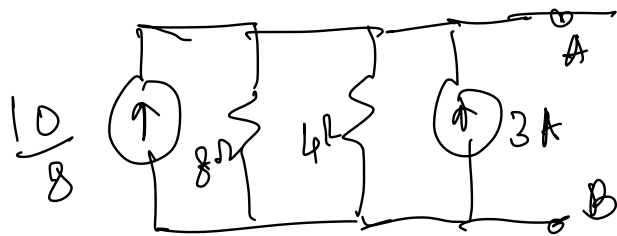
$$I_L = I_{L1} + I_{L2}$$

→ What if R_L changes to 10, 20, 30 and then 100 Ω , do we need to do the calculations again for each R_L ?

What if we reduce the circuit connected to R_L ?

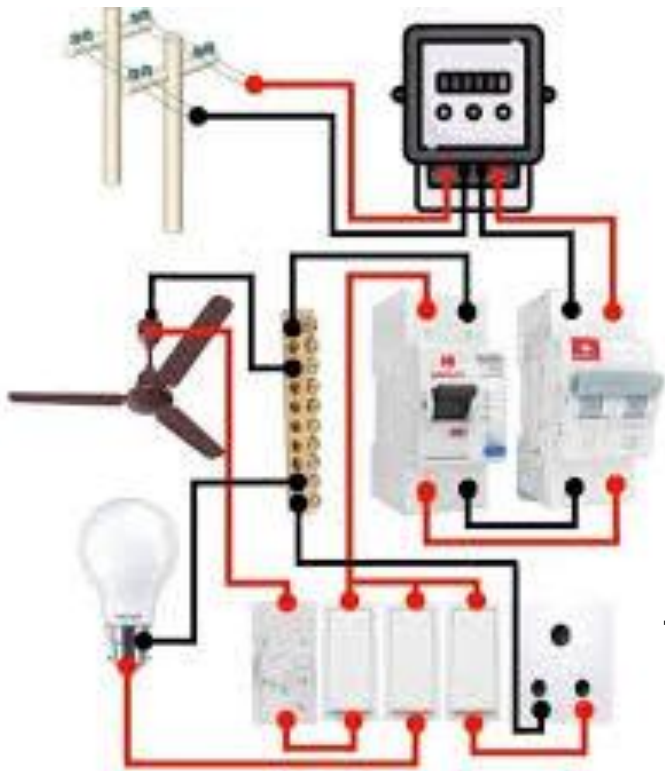


Remember source transformation that we have talked about?

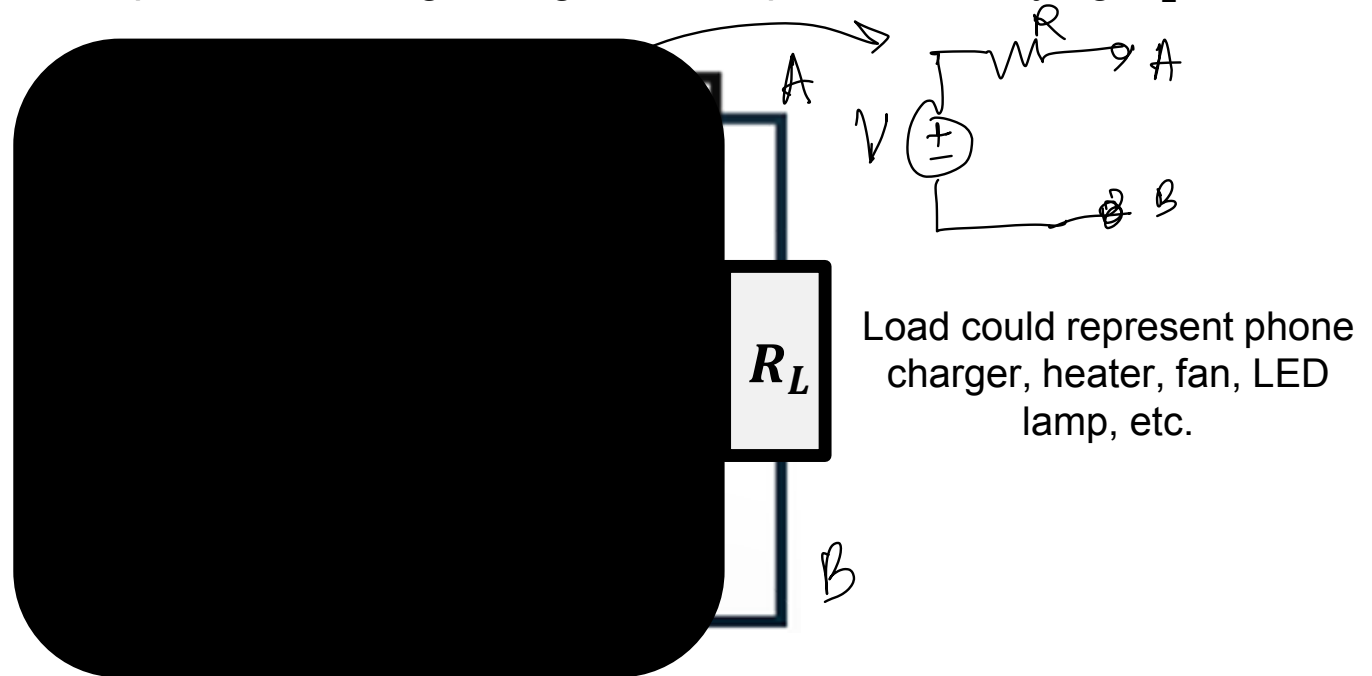


Let us take a practical example with Load

Different load/devices connected to the house wiring.

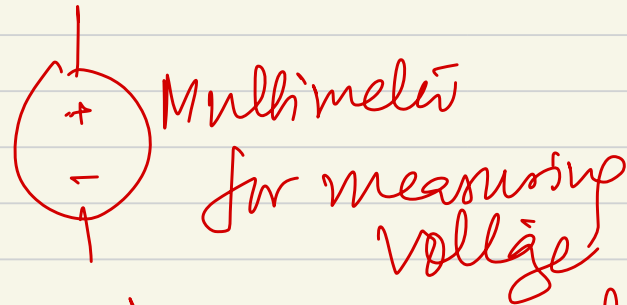
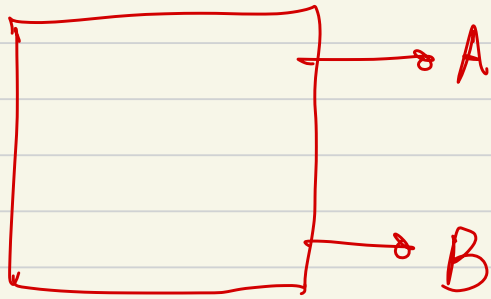


Example: Calculating voltage/current/power for varying R_L .

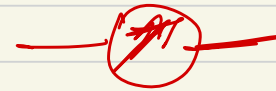
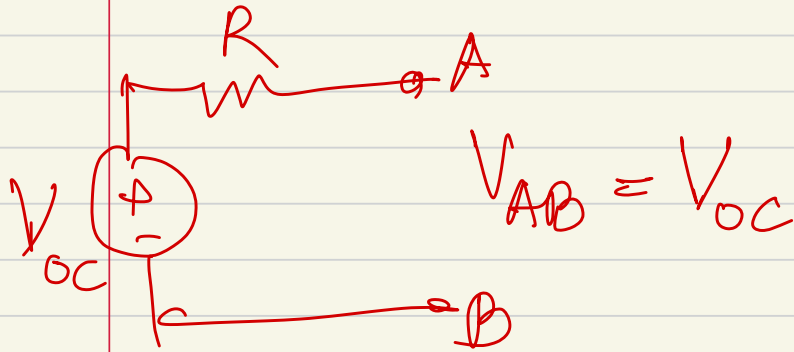


→ This could result in a complex linear circuit.

Q. Can we simplify or convert this complex circuit to a black box representation?

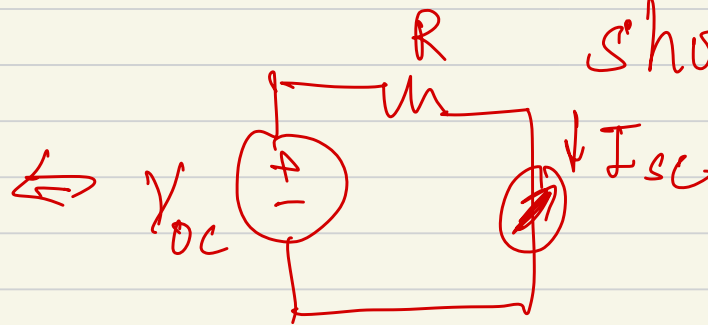
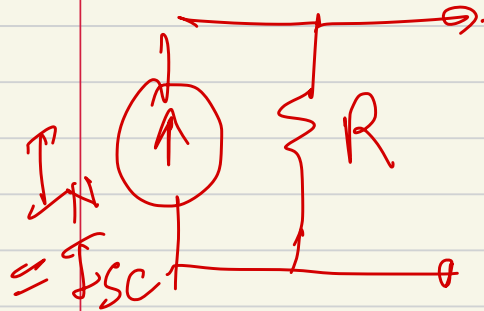


open circuit voltage ($R_L \rightarrow \infty$)



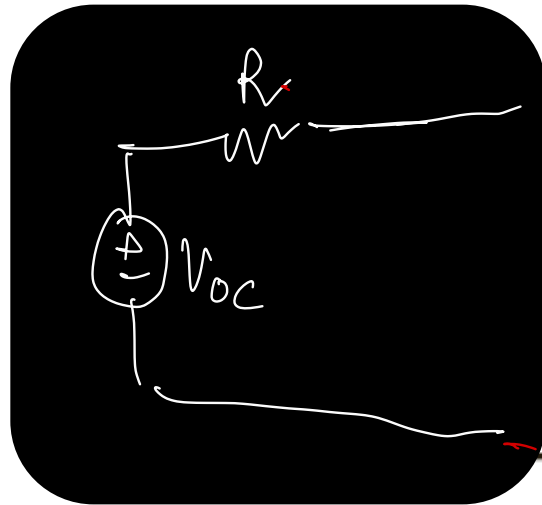
DMM for current

short-circuit current ($R_L \rightarrow 0$)



$$R = \frac{V_{OC}}{I_{SC}}$$

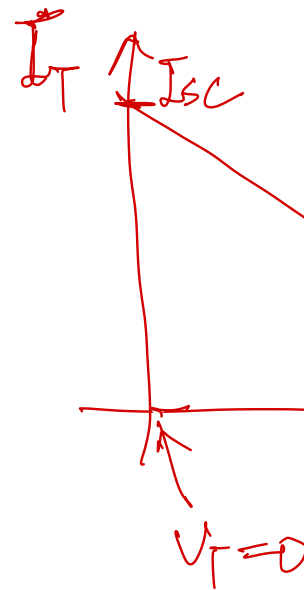
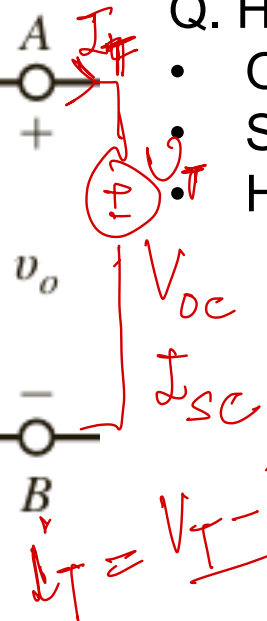
Black Box Representation: Practical View



Q. How do we characterize this block?

- Open Circuit Voltage
- Short-circuit current

How do they relate to each other?

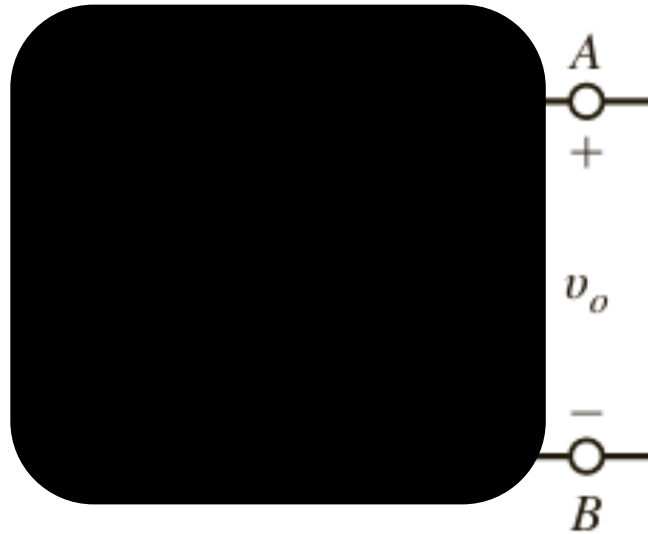


V_T is the test source representing varying load.

I_T is the current drawn by V_T .

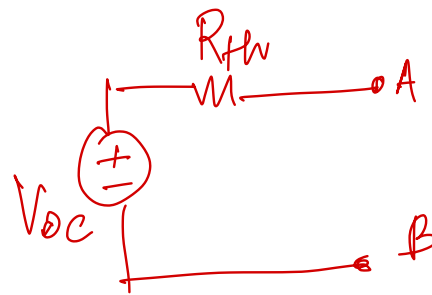
→ this is the signature of a linear circuit.

Black Box Representation: With Voltage/Current Source



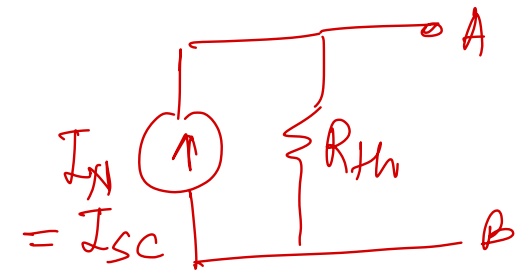
Q. How do we model this block in terms of voltage source & resistor or current source & resistor?

Voltage Source and Resistor



Theremin's Equivalent

Current Source and Resistor

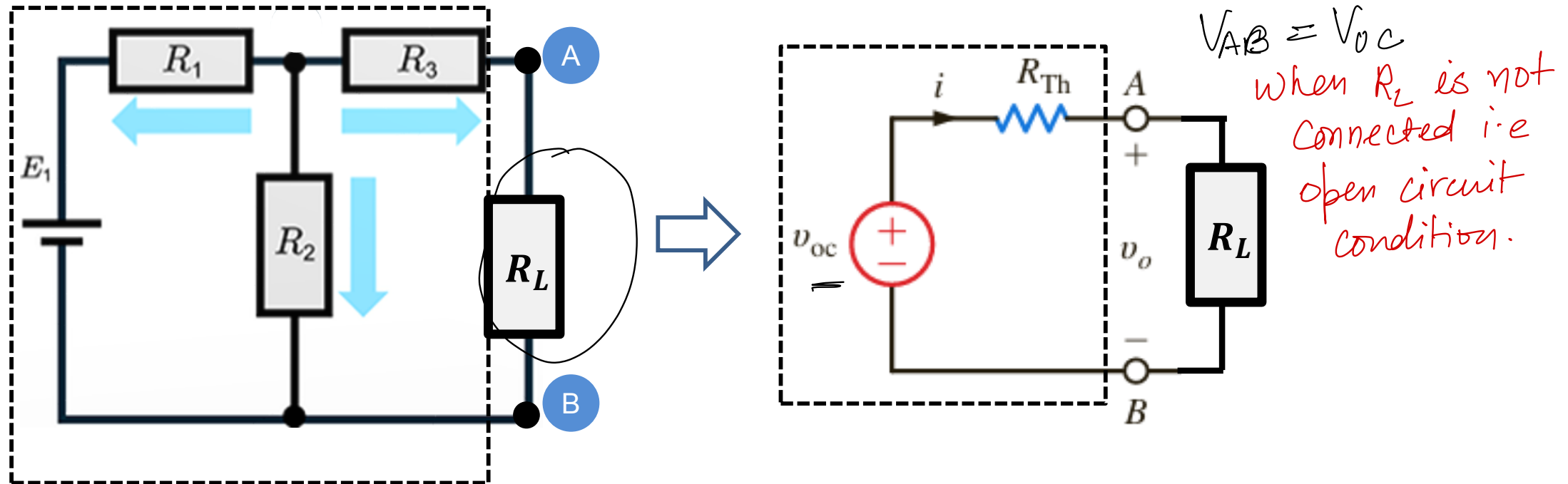


Norton's Equivalent



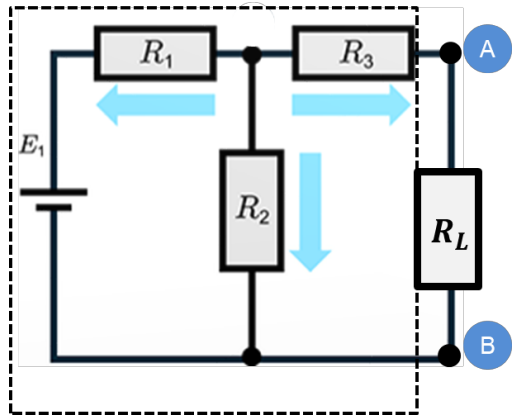
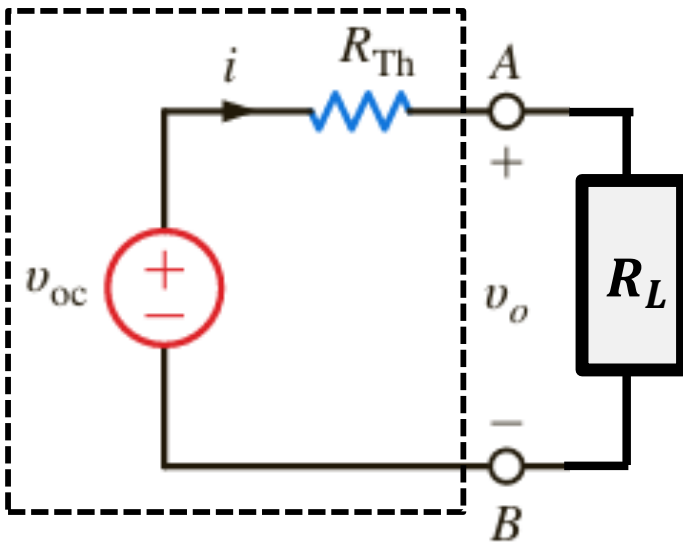
Transforming a network using Thévenin Theorem

Example: Calculating voltage/current/power for varying R_L .



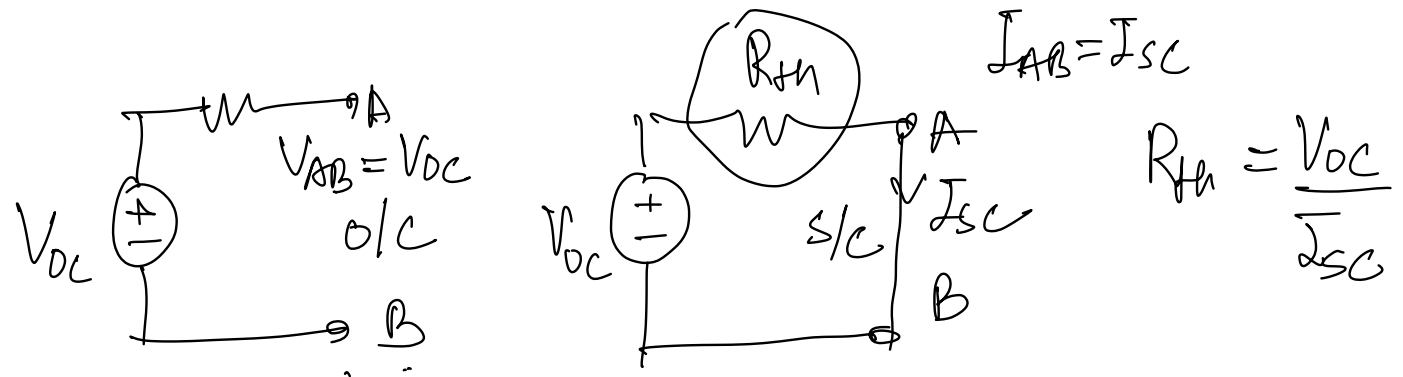
- How do we calculate v_{oc} and R_{th} of the given resistive network between nodes A and B?

Thévenin Theorem

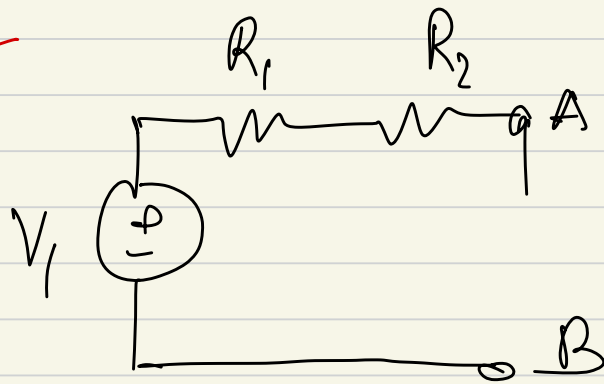


To calculate v_{oc} and R_{Th} , we can follow the following procedure:

- Determine the open-circuit voltage v_{oc} .
- Deactivate the independent sources and find the Thévenin resistance R_{Th} looking back into the terminals.
- Do not deactivate dependent sources.



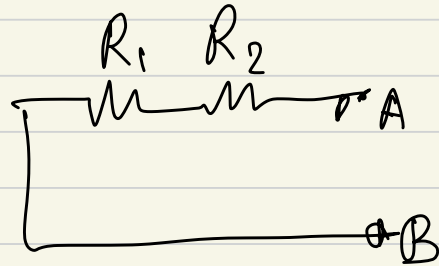
Example 1:-



$$V_{oc} = V_1$$
$$I_{sc} = \frac{V_1}{R_1 + R_2}$$

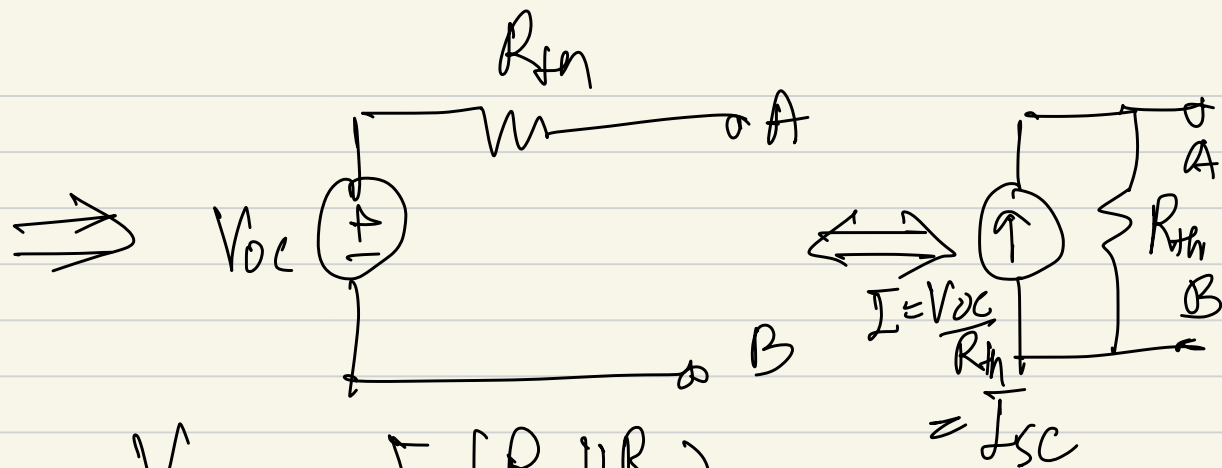
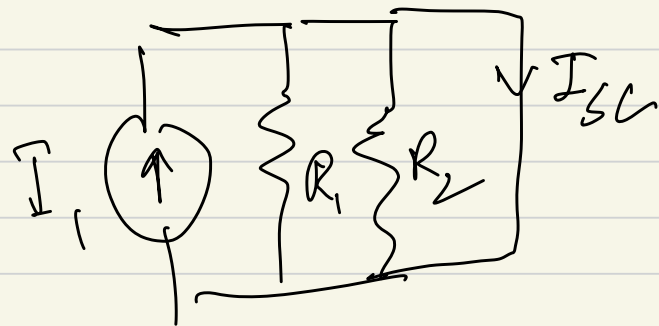
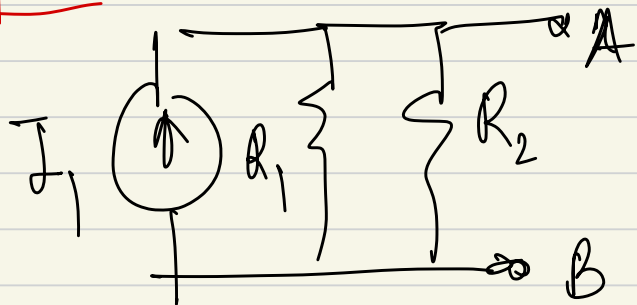
$$R_{Th} = \frac{V_{oc}}{I_{sc}} = R_1 + R_2$$

for R_{Th} , V_1 deactivated



$$R_{AB} = R_{Th} = R_1 + R_2$$

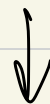
example 2:-



$$V_{oc} = I_1 (R_1 \parallel R_2)$$

$$I_{sc} = I_1$$

$$R_{th} = R_1 \parallel R_2$$

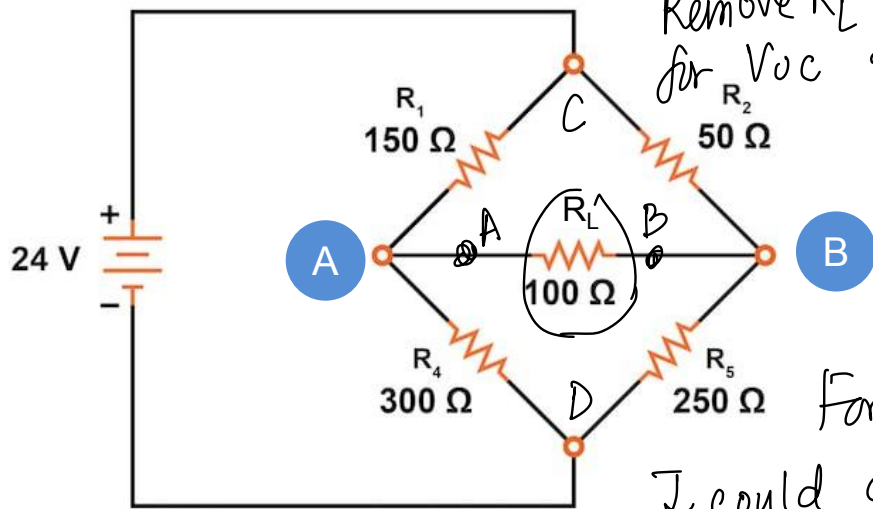


for R_{th} :
deactivating I_1

$$R_1 \parallel R_2 \Rightarrow R_{AB} = R_{th} = R_1 \parallel R_2$$

Application in Unbalanced Wheatstone Bridge

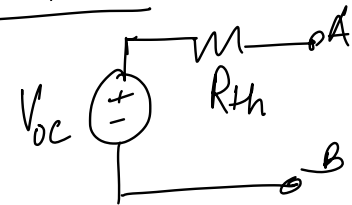
- Find the current flowing through R_L . $\rightarrow$ We can convert the network into thevenin equivalent b/w A & B nodes:-



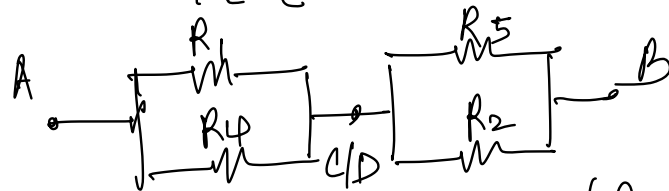
Remove R_L for V_{oc} : $V_{oc} = V_{AB}$

$$= V_A - V_B$$

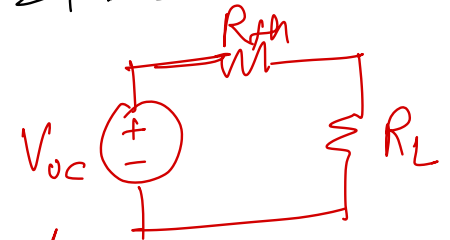
$$= 24 \times \frac{300}{450} - 24 \times \frac{250}{300}$$



For calculating R_{th} , I could calculate I_{sc} i.e. I_{AB} or I could calculate R_{th} by deactivating the 24V source i.e. C & D are shorted



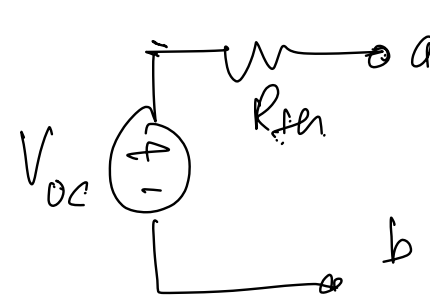
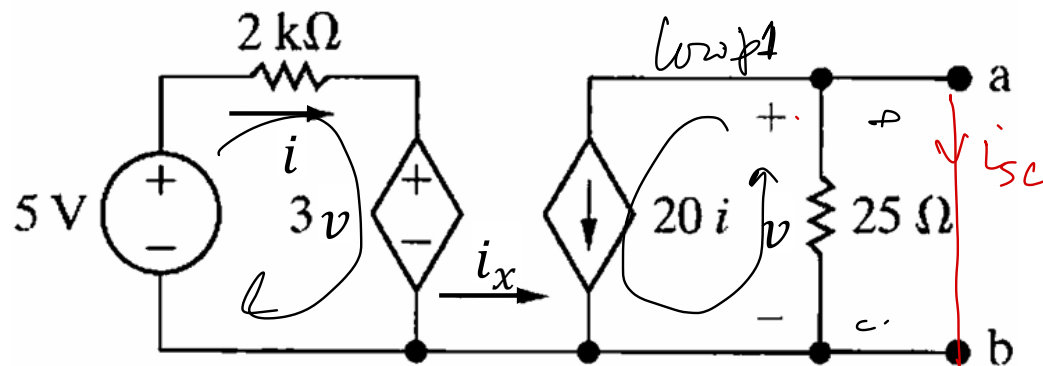
$$\Rightarrow R_{AB} \text{ or } R_{th} = (R_1 \parallel R_4) + (R_2 \parallel R_5)$$



$$\Rightarrow I_L = \frac{V_{oc}}{R_{th} + R_L} =$$

For circuits with dependent sources

- Find the Thevenin equivalent.



$$V_{oc} = -5V$$

$$I_{sc} = -\frac{1}{20}A$$

$$R_{th} = 5 \times 20 = 100\Omega$$

for V_{oc} :

$$V_{oc} = V$$

from loop 1: $V = -20i \times 25$

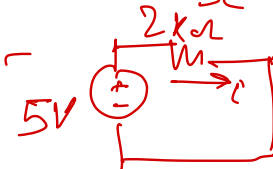
$$V = V_{oc} = -500 \times \frac{5}{500} = -5$$

for R_{th} :

$$R_{th} = \frac{V_{oc}}{I_{sc}}$$

short a & b $\Rightarrow$ 25Ω resistor gone
 $V = 0$
 $I_{sc} = -20i$

for loop 2:



$$i = \frac{5}{2k}, I_{sc} = -20 \times \frac{5}{2k} A$$

Loop 2

for loop 2:

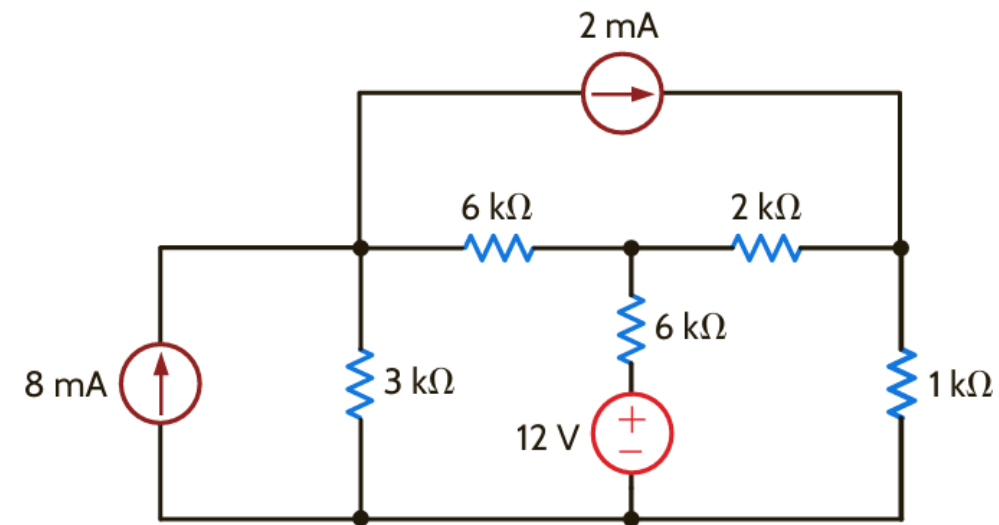
$$5 - 2ki - 3V = 0$$

$$5 - 2000i + 3 \times 500i = 0$$

$$i = \frac{5}{500}$$

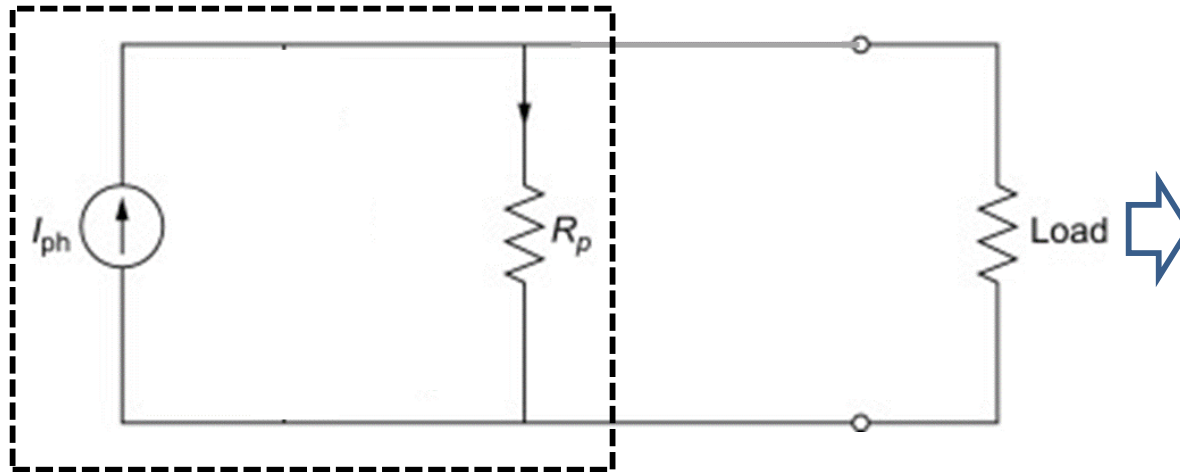
(Take Home) Example

- Find the power supplied by the 12-V source.



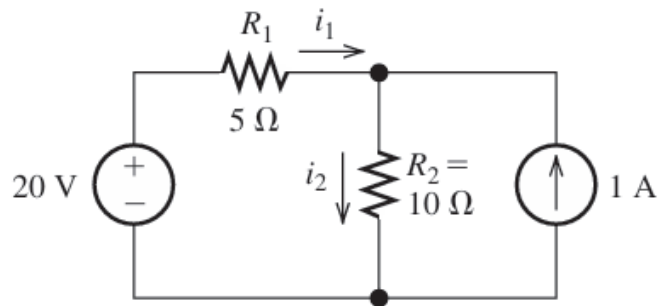
Voltage-Current Dual Representation in a circuit

Photodiode

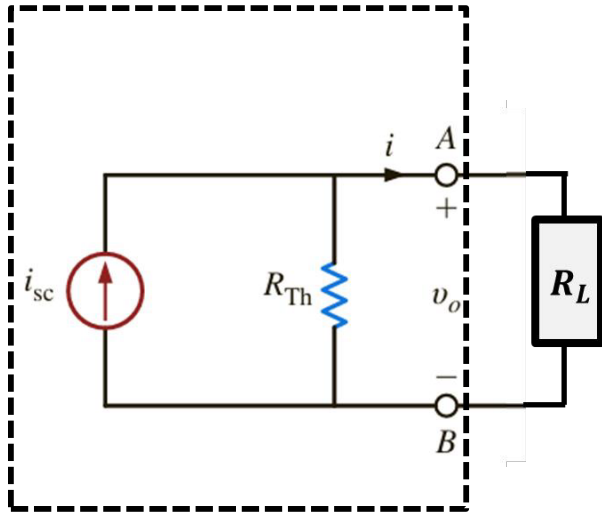


Voltage source equivalent network

Q. Calculate i_1 and i_2 using source transformation.



Norton's Theorem



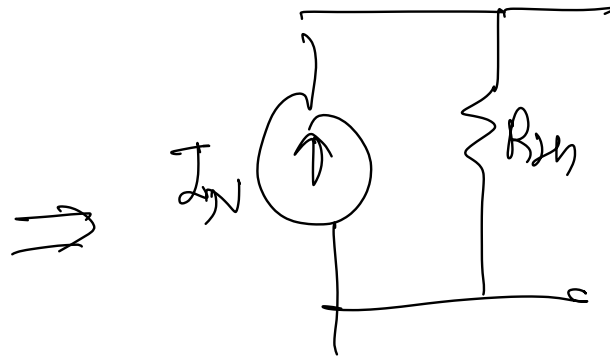
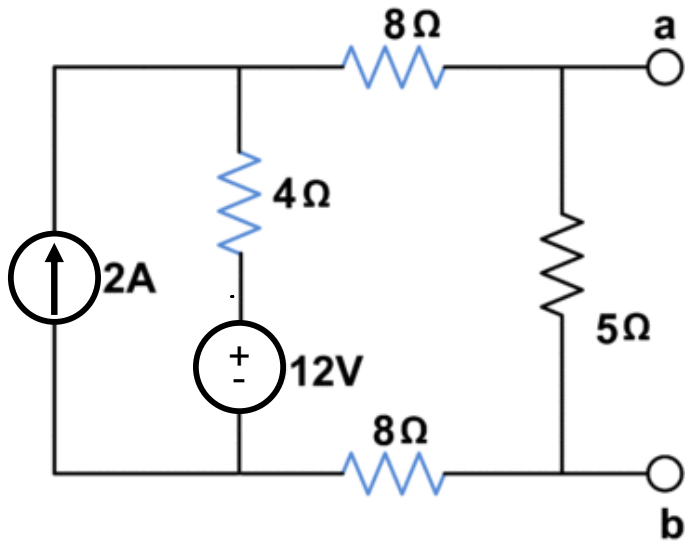
To calculate i_{sc} and R_{Th} , we can follow the following procedure:

- Determine the short-circuit current I_{sc} .
- Deactivate the independent sources and find the Thévenin resistance R_{Th} looking back into the terminals.
- Do not deactivate dependent sources.

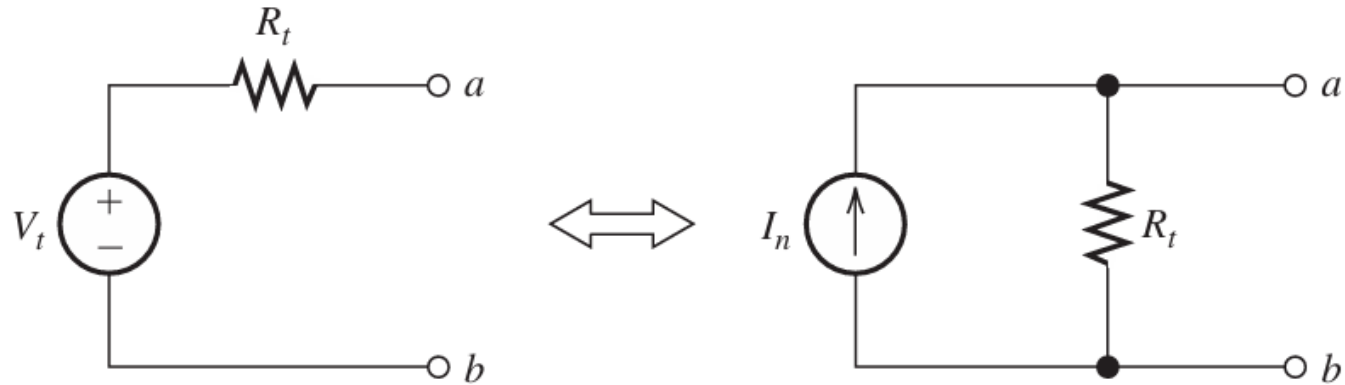
Example

- Draw the Norton Equivalent Circuit.

Home Work



Summary



- Thévenin and Norton theorems let you replace any linear two-terminal network (resistors + voltage/current sources) by a simple equivalent seen from those terminals.
- Thévenin equivalent: one ideal voltage source V_t in series with a resistance R_t .
- Norton equivalent: one ideal current source I_n in parallel with a resistance R_t .
- The two are related by: $V_t = I_n R_t$
- Use them to simplify analysis when the load changes, so you don't re-solve the whole circuit each time.